

A likelihood that carries the recording filter’s state

Draft section for the MacroIR–Bessel paper

1 The measurement problem

The patch-clamp amplifier low-pass filters the current before the digitizer samples it, so the real acquisition kernel is the boxcar composed with the amplifier’s analogue low-pass. In the recordings that motivate this work the filter is a 4-pole Bessel with cutoff $f_c = 10$ kHz and the sampling rate is 50 kHz [1]. A low-pass filter has memory: the value the digitizer records at time t is a weighted average of the recent past of the current $i(t)$,

$$z(t) = \int_0^\infty h(s) i(t-s) ds + \text{filtered instrument noise}, \quad (1)$$

where h is the filter’s impulse response. For a 4-pole Bessel at 10 kHz, h is appreciable for about $100 \mu\text{s}$, which spans five sampling intervals.

The likelihood of the companion paper [2] implements the boxcar exactly: each recorded value is modelled as the plain average of the current over its own acquisition window. What the composition with the analogue filter changes is governed by one dimensionless group, the cutoff frequency against the window, $f_c \Delta$, and the companion paper’s appendix quantifies it: at $f_c \Delta = 0.2$ the white-noise component of a recorded value has 61.5% less variance than the boxcar assigns, and neighbouring recorded values are correlated at 0.641, both effects the boxcar model ignores.

Two separate things go wrong when the kernel spills across window boundaries:

1. a recorded value depends on what the channels did during earlier windows, and the recursion never conditions on that history;
2. consecutive recorded values share signal history and share filtered noise, so they are correlated even once the occupancies at the window boundaries are known, while the recursion treats them as conditionally independent.

On the record of [1] the failure sits exactly where the information is. The single-sample windows around the agonist pulses are $20 \mu\text{s}$ long, so they sit at $f_c \Delta = 0.2$; the $187 \mu\text{s}$ test pulses are only about 2.7 times the filter’s fixed smear of $0.69/f_c = 69 \mu\text{s}$. Grouping samples until the boxcar becomes valid would discard the very windows the ultrashort-pulse experiment exists to produce. A filter-aware likelihood buys those windows back.

The obstruction to fixing this inside the existing recursion is named in the companion paper’s appendix. There, the boundary-state construction is shown equivalent to a filter on the augmented state $(\mathbf{N}, z)$, where z accumulates the window average; the equivalence holds per window, with a reset: the augmented filter must discard the posterior on z and open the next window with $z = 0$, because the next window’s integral is fresh and independent of the old given the occupancy. A filter that remembers across windows forbids exactly that reset. The construction below removes the need for it.

2 The amplifier as part of the model

A k_f -pole filter is itself a small physical system. It has k_f internal variables (the voltages on the capacitors inside the filter), collected in a vector $\boldsymbol{\varphi} \in \mathbb{R}^{k_f}$, which evolve linearly, driven by whatever current enters:

$$\frac{d\boldsymbol{\varphi}}{dt} = \mathbf{A}_f \boldsymbol{\varphi} + \mathbf{b}_f i(t), \quad \text{recorded output} = \mathbf{c}_f^\top \boldsymbol{\varphi}, \quad (2)$$

with $\mathbf{A}_f, \mathbf{b}_f, \mathbf{c}_f$ fixed, known matrices determined by the pole positions and the cutoff. For the 4-pole Bessel, $k_f = 4$ and the transfer function is $H(s) = \theta_4(0)/\theta_4(s/\omega_0)$ with $\theta_4(s) = s^4 + 10s^3 + 45s^2 + 105s + 105$ the reverse Bessel polynomial, realized in companion form, with the frequency scale ω_0 chosen so that the -3 dB point falls at f_c . We write $\nu_1, \dots, \nu_{k_f}$ for the filter poles, with $\text{Re } \nu_p > 0$, so that the eigenvalues of $\mathbf{A}_f$ are $-\nu_p$; the poles come in complex-conjugate pairs and every quantity below recombines to a real number.

The channels are the ensemble of the companion paper: N_{ch} independent Markov channels on K states with generator $\mathbf{Q}$, transition matrix $\mathbf{P}(t) = e^{\mathbf{Q}t}$ (row-stochastic), and conductance vector $\boldsymbol{\gamma}$. Writing $\mathbf{N}(t) \in \mathbb{N}^K$ for the vector of state counts, the current entering the amplifier is

$$i(t) = \boldsymbol{\gamma}^\top \mathbf{N}(t) + i_b + \xi(t), \quad (3)$$

where i_b is the baseline current and ξ is white instrument noise with autocovariance $\mathbb{E}[\xi(t)\xi(t')] = S_0 \delta(t - t')$; S_0 is the noise power spectral density referred to the filter input.

The digitizer does not record $i(t)$. It records the linear readout $\mathbf{c}_f^\top \boldsymbol{\varphi}(t)$ of the filter state at the sampling instants. This observation converts the memory problem into a state problem. The recorded value remembers more about the past than the occupancies at the window boundaries do, which is why no recursion over occupancies alone can be correct on short windows. All of that memory, however, is stored in $\boldsymbol{\varphi}$: since $\boldsymbol{\varphi}$ is driven by i , the joint pair $(\mathbf{N}, \boldsymbol{\varphi})$ is Markov, and its present value determines the statistics of its whole future. Because the filter sits in series with the channels, its states add to the description ($K + k_f$ variables); this is unlike coloured noise modelled as a parallel hidden process, which multiplies the state space.

3 The augmented recursion

3.1 The carried state

The companion paper's recursion carries a Gaussian in the per-channel pair $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with $\boldsymbol{\mu} = \mathbb{E}[\mathbf{N}]/N_{\text{ch}}$ and $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{N})/N_{\text{ch}}$, conditioned on all recorded values so far. The Bessel member carries a Gaussian over the joint of occupancies and filter state, closed at two moments:

$$\boldsymbol{\mu} = \frac{\mathbb{E}[\mathbf{N}]}{N_{\text{ch}}}, \quad \mathbf{m} = \mathbb{E}[\boldsymbol{\varphi}], \quad \boldsymbol{\Sigma} = \frac{\text{Cov}(\mathbf{N})}{N_{\text{ch}}}, \quad \boldsymbol{\Sigma}_{N\boldsymbol{\varphi}} = \frac{\text{Cov}(\mathbf{N}, \boldsymbol{\varphi})}{N_{\text{ch}}}, \quad \boldsymbol{\Sigma}_{\boldsymbol{\varphi}\boldsymbol{\varphi}} = \text{Cov}(\boldsymbol{\varphi}), \quad (4)$$

where the filter blocks $\mathbf{m}$ and $\boldsymbol{\Sigma}_{\boldsymbol{\varphi}\boldsymbol{\varphi}}$ stay at physical scale because $\boldsymbol{\varphi}$ is one physical object, driven by the whole ensemble. $\boldsymbol{\Sigma}_{N\boldsymbol{\varphi}}$ is the structurally new block: it records how uncertainty about where the channels are is tied to uncertainty about the filter's internal state, and it is the conduit through which a recorded value informs the occupancy. Between windows nothing is reset: the quantity whose reset the augmented formulation required, and a filter with memory forbids, is simply carried.

3.2 Prediction

The posterior at the start of a window of length Δ , pushed through the joint dynamics, is the prior at its end. The means propagate through the block-triangular augmented drift,

$$\frac{d}{dt} \begin{pmatrix} \boldsymbol{\mu} \\ \mathbf{m} \end{pmatrix} = \begin{pmatrix} \mathbf{Q}^\top & \mathbf{0} \\ N_{\text{ch}} \mathbf{b}_f \boldsymbol{\gamma}^\top & \mathbf{A}_f \end{pmatrix} \begin{pmatrix} \boldsymbol{\mu} \\ \mathbf{m} \end{pmatrix} + \begin{pmatrix} \mathbf{0} \\ \mathbf{b}_f i_b \end{pmatrix}, \quad (5)$$

so the spectrum of the augmented system is $\text{spec}(\mathbf{Q}) \cup \{-\nu_p\}$: the channel eigenvalues and the filter poles, with no mixing. The covariance blocks propagate by the laws of total expectation and variance over the hidden path, conditioned on the boundary occupancies, exactly as in the companion paper’s boundary-state construction; the only new ingredients are the endpoint-conditioned moments of exponentially weighted conductance integrals, given in Section 4. Every expectation the propagation needs is an affine or bilinear function of the moments already carried in (4), so the moment propagation itself introduces no approximation.

3.3 The two reads

The observation model splits into two cases, according to how the record was produced.

Native-rate window (one raw sample). The recorded value is the filter output at the sampling instant,

$$z = \mathbf{c}_f^\top \boldsymbol{\varphi}(\Delta). \quad (6)$$

There is no fresh observation noise in this read: the instrument noise entered before the filter, and its contribution already lives inside $\boldsymbol{\Sigma}_{\varphi\varphi}$ (Section 4.3).

Grouped window (an average of several raw samples). The recorded value is the average of the filter output over the window. We add one accumulator variable u , the direct heir of the augmented state z of the companion paper’s appendix, now fed by the filter output rather than by the raw current:

$$\frac{du}{dt} = \mathbf{c}_f^\top \boldsymbol{\varphi}, \quad u(0) = 0, \quad z = \frac{u(\Delta)}{\Delta}. \quad (7)$$

The reset of u at the start of each recorded window is a property of the recorder, whose averaging genuinely restarts there, so it remains as legitimate as it was in the boxcar member. The accumulator is an integrator, a pole at zero; it exists only within the window and is discarded after the update. Applying the grouped read to a native-rate sample would impose an averaging the hardware never performed, so the member switches reads per window.

3.4 Update

The recorded value z is a linear functional of the augmented state, so the Bayes update is Gaussian conditioning of the joint moments, with the same three blocks as the companion paper: the covariance of the state, the variance of the recorded value, and the covariance between the state and the recorded value. For the native-rate read,

$$z^{\text{pred}} = \mathbf{c}_f^\top \mathbf{m}, \quad \sigma^2 = \mathbf{c}_f^\top \boldsymbol{\Sigma}_{\varphi\varphi} \mathbf{c}_f + \sigma_{\text{fl}}^2, \quad \mathbf{g}_\varphi = \boldsymbol{\Sigma}_{N\varphi} \mathbf{c}_f, \quad \delta = z - z^{\text{pred}}, \quad (8)$$

where $\mathbf{g}_\varphi$ takes the place of $\mathbf{g} = \boldsymbol{\Sigma}\boldsymbol{\gamma}$ in the companion paper: it is the covariance between the state a channel is in and the value the amplifier is about to hand the digitizer. The term σ_{fl}^2 is an optional

additive floor for noise components with no finite-dimensional linear realisation (Section 4.3). The posterior follows by conditioning:

$$\begin{aligned}\boldsymbol{\mu}^{\text{post}} &= \boldsymbol{\mu} + \frac{\mathbf{g}_\varphi}{\sigma^2} \delta, & \boldsymbol{\Sigma}^{\text{post}} &= \boldsymbol{\Sigma} - \frac{N_{\text{ch}} \mathbf{g}_\varphi \mathbf{g}_\varphi^\top}{\sigma^2}, \\ \mathbf{m}^{\text{post}} &= \mathbf{m} + \frac{\boldsymbol{\Sigma}_{\varphi\varphi} \mathbf{c}_f}{\sigma^2} \delta, & \boldsymbol{\Sigma}_{\varphi\varphi}^{\text{post}} &= \boldsymbol{\Sigma}_{\varphi\varphi} - \frac{(\boldsymbol{\Sigma}_{\varphi\varphi} \mathbf{c}_f)(\boldsymbol{\Sigma}_{\varphi\varphi} \mathbf{c}_f)^\top}{\sigma^2}, \\ & & \boldsymbol{\Sigma}_{N\varphi}^{\text{post}} &= \boldsymbol{\Sigma}_{N\varphi} - \frac{\mathbf{g}_\varphi (\boldsymbol{\Sigma}_{\varphi\varphi} \mathbf{c}_f)^\top}{\sigma^2},\end{aligned}\tag{9}$$

which reduces, block by block, to the companion paper's update when the filter rows are removed. The grouped read conditions on $u(\Delta)/\Delta$ instead, with the same structure. Two things happen here that the boxcar member cannot express. The datum updates the filter state as well as the occupancy: it says where the capacitors are, and that knowledge carries into the next window. And the correlation between neighbouring recorded values is accounted for, because the predictive distribution of each sample conditions on the previous ones through the carried filter moments.

3.5 The likelihood

The log-likelihood is the chain rule over windows,

$$\log p(z_1, \dots, z_T) = \sum_{j=1}^T \log \mathcal{N}(z_j; z_j^{\text{pred}}, \sigma_j^2),\tag{10}$$

with z_j^{pred} and σ_j^2 the predictive mean and variance of sample j given all earlier samples, read off (8) after the prediction step of window j .

4 The interval integrals

4.1 Pole decomposition

The impulse response of the filter is a sum of decaying exponentials, one per pole:

$$h(s) = \mathbf{c}_f^\top e^{\mathbf{A}_f s} \mathbf{b}_f = \sum_{p=1}^{k_f} r_p e^{-\nu_p s},\tag{11}$$

with residues r_p . Solving (2) over one window,

$$\boldsymbol{\varphi}(\Delta) = e^{\mathbf{A}_f \Delta} \boldsymbol{\varphi}(0) + \int_0^\Delta e^{\mathbf{A}_f(\Delta-s)} \mathbf{b}_f i(s) ds,\tag{12}$$

and in the eigenbasis of $\mathbf{A}_f$ the channel part of the integral decomposes, channel by channel, into the scalar quantities

$$W_p = \int_0^\Delta e^{-\nu_p(\Delta-s)} \gamma_{X(s)} ds,\tag{13}$$

where $X(s)$ is the state of one channel at time s . W_p is the channel's conductance history as the filter's p -th mode remembers it. Setting $\nu_p = 0$ recovers the plain time integral of the conductance, which is the boxcar member's object.

4.2 Endpoint-conditioned moments and their spectral evaluation

To propagate the covariance blocks, the recursion needs the moments of W_p conditioned on the channel's occupancy at the window boundaries, in direct analogy with the boundary-conditioned window mean $\bar{\gamma}_{i \rightarrow j}$ and per-state variance of the companion paper:

$$\bar{\gamma}_{i \rightarrow j}^{(p)} = \mathbb{E}[W_p \mid X(0) = i, X(\Delta) = j], \quad \overline{M}_i^{(pq)} = \mathbb{E}[W_p W_q \mid X(0) = i]. \quad (14)$$

The asymmetry in the conditioning is deliberate and it is where the cost is contained. First moments are needed pair-resolved, because the covariance between where a channel ends and what it contributed to the filter is what feeds $\Sigma_{N\varphi}$ and the update vector $\mathbf{g}_\varphi$. Second moments enter only $\Sigma_{\varphi\varphi}$, where the end state is summed over, and a pole-discounted propagator applied to the all-ones vector collapses,

$$e^{(\mathbf{Q} - \nu \mathbf{I})t} \mathbf{1} = e^{-\nu t} \mathbf{1}, \quad (15)$$

so each pole pair (p, q) costs a matrix-vector chain rather than a $K \times K$ table.

All of these integrals are evaluated by the machinery the boxcar member already uses. With the spectral decomposition of the generator, $\mathbf{Q} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^{-1}$ with eigenvalues λ_a , the boxcar member's conditioned moments reduce to divided differences of the exponential,

$$D[x, y] = \frac{e^x - e^y}{x - y}, \quad D[x, x] = e^x, \quad (16)$$

evaluated at pairs of arguments $\lambda_a \Delta$, $\lambda_b \Delta$. An exponential weight in (13) does nothing except shift arguments inside the very same functions: the generic building block is

$$\int_0^\Delta e^{\lambda_a s} e^{(\lambda_b - \nu_p)(\Delta - s)} ds = \Delta D[\lambda_a \Delta, (\lambda_b - \nu_p) \Delta], \quad (17)$$

and second moments involve the analogous second divided differences at argument triples with shifts ν_p , ν_q . No new class of integral appears; the existing code shapes run with shifted inputs, and setting every $\nu_p = 0$ reproduces the boxcar member's formulas identically.

4.3 Instrument noise

The filter that colours the signal also colours the white instrument noise ξ , and its contribution to the covariance of φ over a window has the closed form

$$\Sigma_\xi(\Delta) = S_0 \int_0^\Delta e^{\mathbf{A}_f s} \mathbf{b}_f \mathbf{b}_f^\top e^{\mathbf{A}_f^\top s} ds, \quad (18)$$

which in the eigenbasis of $\mathbf{A}_f$ is again the divided differences of (16), at arguments $-(\nu_p + \nu_q)\Delta$. This term generalises, and replaces, the white per-sample instrumental variance ϵ^2 of the companion paper; the noise correlations it induces across samples are then carried by $\Sigma_{\varphi\varphi}$ like any other filter memory. It also keeps the predictive variance σ^2 in (8) strictly positive, since the pair $(\mathbf{A}_f, \mathbf{b}_f)$ is controllable. Noise components with no finite-dimensional linear realisation, such as $1/f$ noise, stay outside the augmentation as the additive per-observation floor σ_{ff}^2 .

5 The boxcar member as an exact special case

Delete the φ rows and keep only the accumulator u , so that the observation system is $\mathbf{A}_f = 0$, $\mathbf{b}_f = 1$: the augmented state collapses to the $(\mathbf{N}, z)$ of the companion paper’s appendix, and the equations of Section 4 collapse, term by term, to the boxcar member. The same collapse is visible in the algebra as the limit $\nu_p = 0$ of (17). This subcase is the main regression anchor of the implementation: the new code, run in that configuration, must reproduce the existing, validated member to machine precision. The limit $f_c \rightarrow \infty$ also recovers the boxcar member, though only approximately at finite f_c , and serves as a secondary check.

6 What is exact and what is approximate

The moment propagation is exact: every expectation the recursion needs is an affine or bilinear function of the moments it already carries, so no information is silently dropped in the prediction step. What remains approximate is exactly what is approximate in the boxcar member: the Bayes update treats the joint distribution as Gaussian, and between windows only two moments are carried. Both closures now act on a larger object, the joint of occupancy and filter state, so their error is of the same kind but need not be of the same size. We measure it on simulated ground truth rather than assume it; the construction is an extended closure whose accuracy is reported, and we make no exactness claim for the likelihood itself.

7 Cost

Relative to the boxcar member, the growth is a combinatorial factor in how many argument pairs the divided differences are evaluated at, with the modes of the observation system playing the role of extra eigenvalues: first-moment tables grow linearly in the number of modes (one pair-resolved table per pole, plus one for the accumulator), while second-moment quantities grow with the number of pole pairs but are held at matrix-vector cost by (15). Conjugate symmetry halves the complex arithmetic. The overall cost is polynomial in the state count K and the pole count k_f , and independent of the number of channels N_{ch} . Two properties of real records reduce it further: distinct window lengths are few, so the window quantities are memoized on the triple (length, stimulus, pole set); and on windows with $f_c \Delta \gg 1$ the filter has settled well within the window, and the computation reverts to the boxcar member’s cost through an asymptotic branch. On the record of [1], 742 of 1078 windows sit in that regime.

8 Fixed filter parameters and identifiability

The filter’s cutoff and pole count are properties of the recording apparatus. They are fixed at their known values and never fitted: they trade almost degenerately against fast rate constants and against the noise amplitude, so releasing them would launder model error into pseudo-physical parameters. Rates faster than f_c remain weakly identified under any correct treatment, because the filter genuinely destroyed that information before the digitizer saw it; the honest outcome is wider posteriors on those rates, and a narrower posterior there would be a defect of the method, a point the simulations verify. Under a filter the noise amplitude must be stated as the input-referred spectral density S_0 of (3); conventions that quote a variance per sample depend on the bandwidth and do not transfer across members.

9 Relation to existing treatments

Filters inside single-channel likelihoods have a long history. Michalek and colleagues identified the anti-aliasing filter as a cause of wrong model identification and modelled it with moving-average filtered hidden Markov models [3]; Venkataramanan, Kuc and Sigworth built the metastate formulation, expanding the hidden state over the filter’s memory so that band-limited sampled data become Markov again [4]; Qin, Auerbach and Sachs modelled the filter as a finite impulse response within a hidden Markov likelihood and describe an extension to multiple channels [5]. These treatments are discrete-time and truncate the filter to a sampled impulse response, with a cost that grows exponentially in the filter memory expressed in states per channel, which confines them to one or few channels. Absorbing a linear filter into an augmented state is textbook filtering theory [6]; propagating integrated or averaged observations through an augmented state with matrix-exponential machinery goes back at least to Zdrozny [7], has a general integral primitive in Carbonell, Jiménez and Pedroso [8], and a recent instance with overlapping windows in Rubenzahl and colleagues [9].

What this member contributes is the combination needed for macroscopic channel data: the endpoint-conditioned mean, variance and cross-sample covariance of a Bessel-filtered current for N_{ch} independent channels, computed from the generator’s spectral decomposition at a cost polynomial in state count and pole count and independent of N_{ch} , exact for the analogue kernel where the treatments above are exact only for a sampled truncation; and, with it, the ability to fit a macroscopic record at its native sampling rate, including the windows the filter smears.

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